Intro to Machine Learning

[Kaggle의 Housing Price 데이터 활용]

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Inspien / Dev Team

April 16, 2026

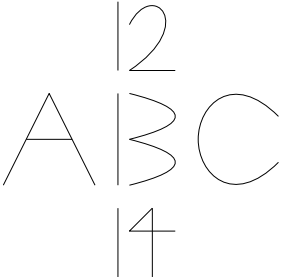
Outline

1. Why ML?
2. Kaggle?
3. Experiences
4. Insights

Complex Problems

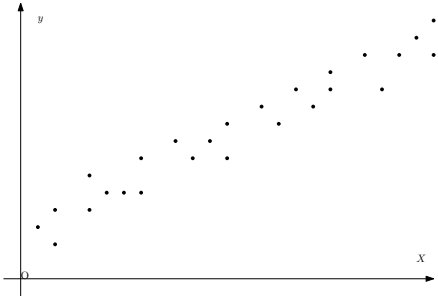
(a)

(b)



(c)

Simple Puzzle (with linear regression)



(a) Simple Problem

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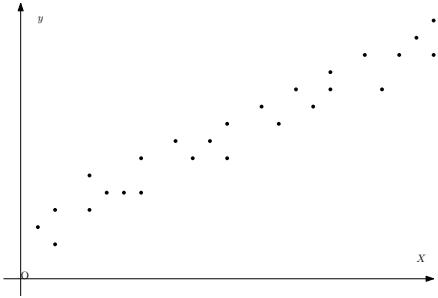

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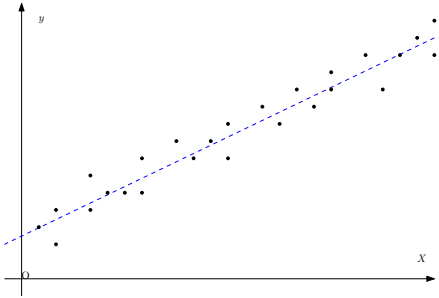
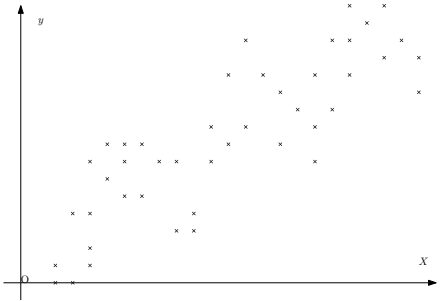


(b) Linear Regression (Simple)

Complicated Puzzle (with Machine Learning)



(a) Complicated Problem

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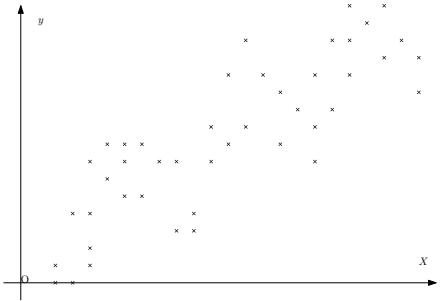

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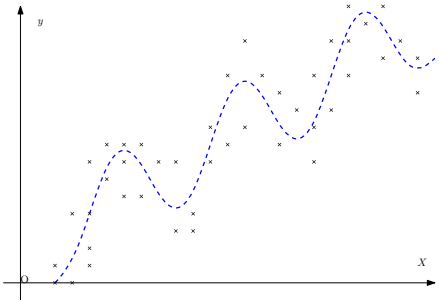

(b) Machine Learning (Complicated)

Kaggle

Kaggle

- Dataset
- ML Problems
- ‘Kaggle Notebook’

Courses I took:

- Intro to Machine Learning (7)
- Pandas (6)
- Intermediate ML (7)

Problem

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집의 여러 정보를 기반으로 집값을 예측하는 모델 만들기.

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- 데이터 (X, y): 1168개
- X 값 (columns): 36가지
  $(MasVnrArea, GarageYrBlt, LotFrontage, \ldots)$
- y 값 : 가격

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LotFrontage	212
MasVnrArea	6
GarageYrBlt	58

Table: number of missing values

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Core Problem

How do I handle the missing data?

Imputing vs. Dropping

idx	*MVA*	*GYB*	*LF*
1	196	2003	65
2	0	1976	80
3	162	2001	68
4	0	**NA**	60
5	350	2000	84

Table: Original Data

Imputing vs. Dropping

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Table: Drop GYB

Experiment

Exp #	*MasVnrArea*	*GarageYrBlt*	*LotFrontage*	MSE (Error) [$]
Baseline 1	DROP	DROP	DROP	17837.82570776256
Baseline 2	IMP	IMP	IMP	18062.894611872147

Table: Comparison of methods.

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Exp 1	IMP	IMP	DROP	18105.497933789953
Exp 2	IMP	DROP	IMP	**17809.983767123285**

Table: Comparison of methods.

Insights

1. Domain Knowledge: Powerful
2. Segregation via Importance
3. ML $\Rightarrow$intuition